

Student Number:



ROSEVILLE COLLEGE

MATHEMATICS EXTENSION 1

TRIAL EXAMINATION 2004

Time allowed: 2 hours + 5 minutes reading time

Directions

- **Attempt ALL questions.**
- **Show all necessary working, marks may be deducted for careless or untidy work.**
- **Board-approved calculators may be used.**
- **Additional Answer Pages are available.**
- **A table of standard integrals is provided**

QUESTION 1 (*Start a new page*)**MARKS**

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{5x}$. 1
- (b) Evaluate $\int_0^{\sqrt{2}} \frac{dx}{x^2 + 2}$. 3
- (c) If $P(x) = x^4 - 3x^3 + ax^2 - 12$ is divisible by $(x - 3)$, find the value of a . 2
- (d) Use the table of standard integrals to evaluate $\int_3^5 \frac{dx}{\sqrt{x^2 - 4}}$. 2
- (e) Show that $\tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1 + x^2}}$ 2
- (f) Seven people want to sit around a circular table. Two of them want to sit together. In how many ways can the table be arranged? 2

QUESTION 2 (*Start a new page*)**MARKS**

- (a) How many 7 letter words can be made from the word ADDIDAS? 2
- (b) Find the coefficient of x^{12} in the expansion of $\left(\frac{1}{x^2} + x^3\right)^{14}$. 3
- ✓ (c) Find $\frac{d}{dx}\left(\frac{\tan^2 x}{x}\right)$. 2
- (d) (i) Express $\cos \theta - \sqrt{3} \sin \theta$ in the form $A \cos (\theta + \alpha)$ where A and α are constants, $A > 0$ and α is in radians. 2
- (ii) Hence solve the equation $\cos \theta - \sqrt{3} \sin \theta = \sqrt{3}$ for $-\pi \leq \theta \leq \pi$. 3

QUESTION 3 (Start a new page)

MARKS

- (a) A circle is expanding so that the rate of increase in its radius is 0.75 cm s^{-1} .

Find the rate of increase in its area when

- (i) the radius is 10cm
 (ii) the circumference is 10cm.

2

2

- (b) Solve $3^{2x} = 5$.

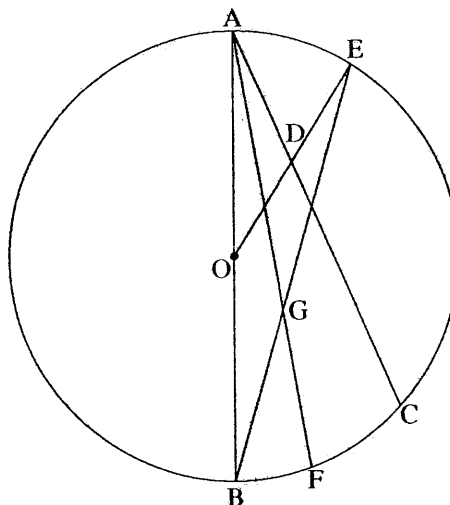
2

- (c) The equation $2x^3 + 2x^2 + 4x + 1 = 0$ has roots α, β, γ . Find the value of

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

3

- (d) In the figure, AOB is the diameter of a circle centre O . D is a point on chord AC such that $DA = DO$ and OD is produced to E . AF is the bisector of $\angle BAC$ and cuts BE in G .
 Prove that $GA = GB$



3

QUESTION 4 (*Start a new page*)**MARKS**

- (a) Use mathematical induction to prove that, for all positive integers
- n
- ,

4

$$\sum_{r=1}^n \frac{r^2}{(2r-1)(2r+1)} = \frac{n(n+1)}{2(2n+1)}$$

- (b) (i) Prove the identity

2

$$\frac{\sin 2\theta}{2 \sin \theta} - \cos \theta \cos 2\theta = 2 \cos \theta \sin^2 \theta$$

- (ii) Hence solve the equation

3

$$\frac{\sin 2\theta}{2 \sin \theta} - \cos \theta \cos 2\theta = \cos \theta \text{ for } 0 \leq \theta \leq 2\pi.$$

- (c) Consider the function
- $f(x) = \cos^{-1}(x-1)$

- (i) Find the domain of the function.

1

- (ii) Sketch the graph of the curve
- $y = f(x)$
- showing clearly the coordinates of the endpoints

2

QUESTION 5 (Start a new page)

MARKS

- (a) Corn cobs are cooked by immersing them in boiling water. On being removed, a corn cob cools in the air according to the equation

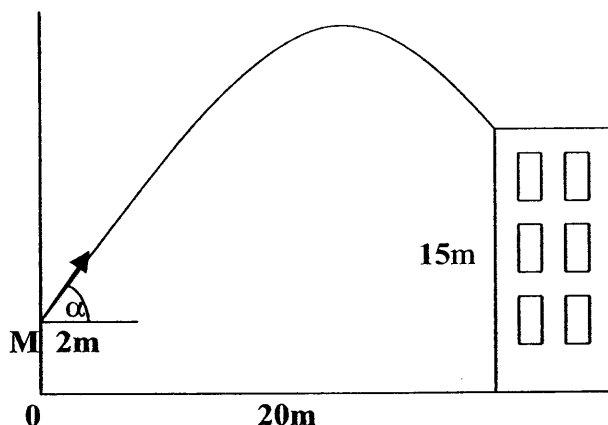
$$\frac{dT}{dt} = -k(T - T_0) \text{ where } t \text{ is time in minutes, } T \text{ is temperature in } ^\circ\text{C} \text{ and } T_0 \text{ is the temperature of the air, while } k \text{ is a positive constant.}$$

- (i) Verify that $T = T_0 + Ae^{-kt}$ is a solution of the above equation where A is a constant. 2

- (ii) If the temperature of the boiling water is 100°C and that of the air is a constant 25°C , find the values of A and k if a corn cob cools to 70°C in 3 minutes. 2

- (iii) How long should a person wait to enjoy the food at a temperature of 50°C ? 2

- (b) A girl of height 2 metres throws a ball from M to the roof of a 15 metre high building. She throws the ball at an initial velocity of 25m/s , and she is 20m from the base of the building.



- (i) Derive the equations of motion of the ball 3

(Assume $\ddot{x} = 0$ and $\ddot{y} = -10$)

- (ii) Between which two angles of projection must she throw the ball to ensure that it lands on the roof of the building? 3

QUESTION 6 (Start a new page)

MARKS

(a) Let $f(x) = \tan^{-1}\left(\frac{2}{x}\right) - \tan^{-1}(-2x)$ where $x > 0$.

Find $f'(x)$.

3

- (b) A particle is moving in a straight line. At time t seconds, it has displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$. The particle starts from O and you are given that $v = (2 - x)^2$

(i) Find an expression for a in terms of x

2

(ii) Find an expression for x in terms of t

3

(iii) Find the distance from O when the particle has a speed of 1 ms^{-1} .

2

- (c) At a football club a team of 13 players is to be chosen from a pool of 32 players consisting of 20 Australian-born players and 12 players born elsewhere. What is the probability that the team will consist of all Australian-born players?

2

QUESTION 7 (*Start a new page*)**MARKS**

(a) (i) Express $\cos 2A$ in terms of $\sin^2 A$ 1

(ii) Hence find the exact value of $\int_0^{\pi} \sin^2 \frac{x}{4} dx$ 2

(b) A particle moves along a straight line such that its displacement x m from an origin O at time t seconds is given by

$$x = 4 \sin \frac{\pi}{2} t$$

(i) Show that this is Simple Harmonic Motion 2

(ii) State the amplitude and period of this motion 1

(iii) Calculate the maximum speed attained by this particle 1

(c) Suppose that $(5 + 2x)^{12} = \sum_{k=1}^{12} a_k x^k$

(i) Use the binomial theorem to write an expression for a_k 2

(iii) Show that $\frac{a_{k+1}}{a_k} = \frac{24-2k}{5k+5}$ 3

End of paper

Question (1)

$$(a) \lim_{x \rightarrow 0} \frac{\sin 4x}{5x} = \frac{4x}{5x} = \frac{4}{5} \rightarrow \textcircled{1}$$

$$(b) \int_0^{\sqrt{2}} \frac{dx}{x^2+2} = \left[\frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} \right]_0^{\sqrt{2}} \rightarrow \textcircled{1}$$

$$= \frac{1}{\sqrt{2}} [\tan^{-1} 1 - \tan^{-1} 0] \rightarrow \textcircled{1}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\pi}{4}$$

$$= \frac{\pi}{4\sqrt{2}} \rightarrow \textcircled{1}$$

$$(c) P(x) = x^4 - 3x^3 + ax^2 - 12$$

$$P(3) = 0$$

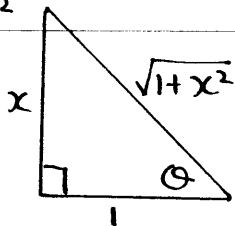
$$0 = 81 - 81 + 9a - 12 \rightarrow \textcircled{1}$$

$$\therefore a = \frac{4}{3} \rightarrow \textcircled{1}$$

$$(d) \int_3^5 \frac{dx}{\sqrt{x^2-4}} = \left\{ \ln(x + \sqrt{x^2-4}) \right\}_3^5 \rightarrow \textcircled{1}$$

$$= \ln(5 + \sqrt{21}) - \ln(3 + \sqrt{5})$$

$$= \ln \frac{5 + \sqrt{21}}{3 + \sqrt{5}} \rightarrow \textcircled{1}$$

$$(e) \tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$$


$$\left. \begin{aligned} \text{let } \theta &= \tan^{-1} x \\ \therefore \tan \theta &= x \\ \therefore \cos \theta &= \frac{1}{\sqrt{1+x^2}} \end{aligned} \right\} \textcircled{1}$$

$$\therefore \theta = \cos^{-1} \frac{1}{\sqrt{1+x^2}} \rightarrow \textcircled{1}$$

$$\therefore \tan^{-1} x = \cos^{-1} \frac{1}{\sqrt{1+x^2}} \rightarrow \textcircled{1}$$

$$(f) \begin{array}{c} \textcircled{X} \textcircled{X} \\ X \quad X \\ X \quad X \end{array} \text{ As if 6 in a circle}$$

$$\therefore \text{No ways to sit together}$$

$$= 5! \times 2 \rightarrow \textcircled{1}$$

$$= 240 \rightarrow \textcircled{1}$$

Question (2)

$$(a) \frac{7!}{3! \times 2!} \rightarrow \textcircled{1}$$

$$= 420$$

$$(b) \left(\frac{1}{x^2} + x^3 \right)^{14}$$

want k such that

$$\left(\frac{1}{x^2} \right)^{14-k} (x^3)^k = x^{12} \rightarrow \textcircled{1}$$

or similar

$$\text{i.e. } 3k - 2(14-k) = 12$$

$$3k - 28 + 2k = 12$$

$$5k = 40$$

$$k = 8 \rightarrow \textcircled{1}$$

$$\therefore \text{Coefficient is } \frac{14!}{8!6!} = 3003$$

$$(c) \frac{d}{dx} \frac{\tan^2 x}{x} = \frac{\textcircled{1} \text{ derivative } \tan^2 x \cdot \textcircled{1} \text{ Quotient rule}}{x^2}$$

$$= \frac{2 \tan x \cdot \sec^2 x \cdot x - \tan^2 x}{x^2}$$

$$(d) \cos \theta - \sqrt{3} \sin \theta = A \cos(\theta + \alpha)$$

$$\text{LHS} = 2 \cos(\theta + \alpha)$$

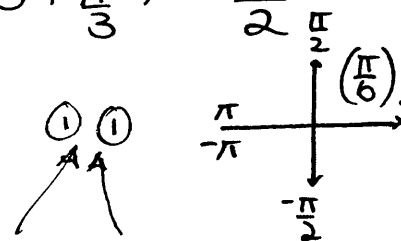
$$\text{Now } \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3} \rightarrow \textcircled{1}$$

$$\therefore \text{LHS} = 2 \cos\left(\theta + \frac{\pi}{3}\right) \rightarrow \textcircled{1}$$

$$(ii) 2 \cos\left(\theta + \frac{\pi}{3}\right) = \sqrt{3}$$

$$\cos\left(\theta + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$



$$\theta + \frac{\pi}{3} = -\frac{\pi}{6}, \frac{\pi}{6}$$

$$\theta = -\frac{\pi}{2}, -\frac{\pi}{6}$$

$$\rightarrow \textcircled{1}$$

Question (3)

(a) $\frac{dr}{dt} = 0.75$

$A = \pi r^2$
 $\frac{dA}{dr} = 2\pi r$

(i) $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$
 $= 2\pi r \times 0.75 \rightarrow \textcircled{1}$

for $r=10 = 2 \times \pi \times 10 \times 0.75$
 $= 15\pi \text{ cm}^2/\text{s} \quad \textcircled{1}$

(ii) $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$

$\textcircled{1} \rightarrow = 2\sqrt{A\pi} \times 0.75$

when $A=10$

$\frac{dA}{dt} = 2\sqrt{10} \times 0.75$
 $= 1.5\sqrt{10} \sqrt{A} \text{ cm}^2/\text{s} \quad \textcircled{1}$

(b) $3^{2x} = 5$

$\ln 3^{2x} = \ln 5$
 $2x \ln 3 = \ln 5$
 $2x = \frac{\ln 5}{\ln 3} \quad \textcircled{1}$

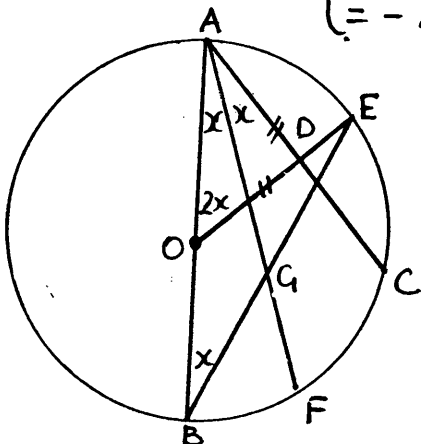
$x = \frac{1}{2} \frac{\ln 5}{\ln 3} \quad \textcircled{1}$
 $x = 0.73248676$

(c) $2x^3 + 2x^2 + 4x + 1 = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$\textcircled{2} \rightarrow \left\{ \begin{aligned} &= (2) \div -\frac{1}{2} \quad \textcircled{1} \\ &= -4 \end{aligned} \right.$

(d)



let $\angle OAF = x = \angle CAD$ (given)
 $\therefore \triangle ADO$ is isosceles ($AO = DO$)
 $\therefore \angle ADO = \angle OAD = 2x$ (equal sides of isos. $\triangle$)

$\therefore \angle ABE = x$ (angle at centre is double $\angle$ at circ)
In $\triangle AGB$
 $\therefore \angle GAB = \angle ABG$ (both x)
 $\therefore \triangle AGB$ is isosceles
 $\therefore GA = GB$ (equal sides of isos. $\triangle$)

Question (4)

$\sum_{r=1}^n \frac{r^2}{(2r-1)(2r+1)} = \frac{n(n+1)}{2(2n+1)}$

$\frac{1}{3} + \frac{4}{15} + \dots + \frac{n^2}{(2n-1)(2n+1)} = \frac{n(n+1)}{2(2n+1)}$

Step $\textcircled{1}$ Prove for $n=1$

when $n=1$
 $\frac{1}{3} = \frac{1(2)}{2(3)}$

$\therefore \text{LHS} = \text{RHS}$

True for $n=1$

Step (2) Assume true for $n=k$

$\therefore \frac{1}{3} + \dots + \frac{k^2}{(2k-1)(2k+1)} = \frac{k(k+1)}{2(2k+1)} \quad \textcircled{1}$

Step $\textcircled{3}$ Prove for $n=k+1$

To Prove

$\frac{1}{3} + \dots + \frac{k^2}{(2k-1)(2k+1)} + \frac{(k+1)^2}{(2k+1)(2k+3)}$

$= \frac{(k+1)(k+2)}{2(2k+3)}$

LHS = $\frac{k(k+1)}{2(2k+1)} + \frac{(k+1)^2}{(2k+1)(2k+3)}$

$= \frac{k(k+1)(2k+3) + (k+1)^2 \cdot 2}{2(2k+1)(2k+3)}$

$= \frac{(k+1)(k(2k+3) + 2(k+1))}{2(2k+1)(2k+3)}$

$$= \frac{(k+1)(2k^2+3k+2k+2)}{2(2k+1)(2k+3)}$$

(1/2)

$$= \frac{(k+1)(2k+5k+2)}{2(2k+1)(2k+3)}$$

$$= \frac{(k+1)(\cancel{2k+1})(k+2)}{2(\cancel{2k+1})(2k+3)}$$

$$= \frac{(k+1)(k+2)}{2(2k+3)}$$

(1)

= RHS

True for $n=k+1$ when true for $n=k$
Step (4)

Since true for $n=1$ from step (3) it's true for $n=2$ and hence $n=3$ and so on for all n . (1/2)

(b) (i) $\frac{\sin 2\theta}{2\sin\theta} - \cos\theta \cos 2\theta = 2\cos\theta \sin^2\theta$

LHS = $\frac{2\sin\theta \cos\theta}{2\sin\theta} - \cos\theta(\cos^2\theta - \sin^2\theta)$

$$\begin{aligned} &= \cos\theta - \cos\theta(1 - 2\sin^2\theta) \\ &= \cos\theta(1 - 1 + 2\sin^2\theta) \\ &= 2\cos\theta \sin^2\theta \\ &= \text{RHS.} \end{aligned}$$

(ii) $\therefore 2\cos\theta \sin^2\theta = \cos\theta$

$$2\cos\theta \sin^2\theta - \cos\theta = 0$$

$$\cos\theta(2\sin^2\theta - 1) = 0 \quad \text{--- (1)}$$

$$\therefore \cos\theta = 0 \quad \sin^2\theta = \frac{1}{2}$$

$$\sin\theta = \pm \frac{1}{\sqrt{2}}$$

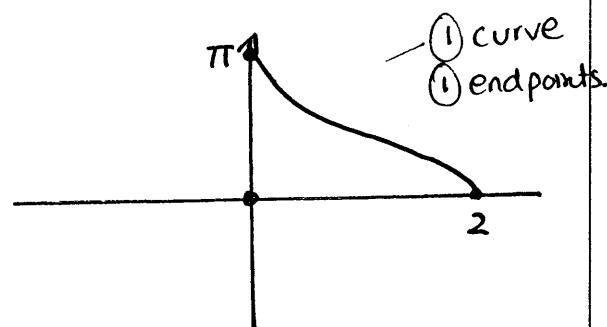
$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}, \dots$$

(2)

(c) $f(x) = \cos^{-1}(x-1)$

(i) domain $-1 \leq x-1 \leq 1$
 $0 \leq x \leq 2$ (1)

(ii)



Question (5)

(a) $\frac{dT}{dt} = -k(T - T_0)$

(i) $T = T_0 + Ae^{-kt}$

$\frac{dT}{dt} = -k \cdot Ae^{-kt}$

$= -k[T - T_0]$

$\therefore$ satisfies equation.

(ii) when $t=0$ $T=100$, $T_0=25$

$\therefore 100 = 25 + A$

$A = 75$

$\therefore T = 25 + 75e^{-kt}$

when $t=3$, $T=70$

$70 = 25 + 75e^{-3k}$

$45 = 75e^{-3k}$

$-3k = \ln\left(\frac{45}{75}\right)$

$k = 0.170275207$

(ii) when $T=50$

$50 = 25 + 75e^{-kt}$

$25 = 75e^{-kt}$

$-kt = \ln\left(\frac{25}{75}\right)$

$t = 6.45 \text{ mins}$
 7 mins

(b) (i) $\ddot{x} = 0$

$$\left. \begin{aligned} \dot{x} &= 25 \cos \alpha \\ \dot{y} &= 25t \cos \alpha \end{aligned} \right\} \textcircled{1}$$

$$\ddot{y} = -10 \quad (t=0 \quad \dot{y} = v \sin \alpha)$$

$$\dot{y} = -10t + 25 \sin \alpha \quad \textcircled{1}$$

$$y = -5t^2 + 25t \sin \alpha + C \quad \begin{cases} t=0 \\ y=2 \end{cases}$$

$$y = -5t^2 + 25t \sin \alpha + 2 \quad \textcircled{1}$$

(ii) $t = \frac{x}{25 \cos \alpha}$ sub m to y

$$y = \frac{-5 \cdot x^2}{25^2 \cos^2 \alpha} + \frac{x \sin \alpha \cdot 25}{25 \cos \alpha} + 2$$

$$y = \frac{-x^2}{125} (\sec^2 \alpha) + x \tan \alpha + 2$$

$$y = \frac{x^2(1 + \tan^2 \alpha) + x \tan \alpha + 2}{125} \quad \textcircled{1}$$

sub m point (20, 15)

$$15 = \frac{-400}{125} (1 + \tan^2 \alpha) + 20 \tan \alpha + 2 \quad \textcircled{1}$$

$$1875 = -400(1 + \tan^2 \alpha) + 2500 \tan \alpha + 250$$

$$1625 = -400 - 400 \tan^2 \alpha + 2500 \tan \alpha$$

$$0 = -400 \tan^2 \alpha + 2500 \tan \alpha - 2025$$

$$0 = 16 \tan^2 \alpha - 100 \tan \alpha + 81$$

$$\tan \alpha = \frac{+100 \pm \sqrt{100^2 - 4 \times 81 \times 16}}{32}$$

$$= \frac{+100 \pm \sqrt{4816}}{32} \quad \textcircled{1}$$

$$\alpha = 79^\circ \text{ and } 44^\circ$$

$$44^\circ \leq \alpha \leq 79^\circ$$

Question (6) -

(a) $f(x) = \tan^{-1}\left(\frac{2}{x}\right) - \tan^{-1}(-2x)$

$$f'(x) = \tan^{-1}\left(\frac{2}{x}\right) + \tan^{-1}(2x) \quad \textcircled{1}$$

$$f'(x) = \frac{-2x^{-2}}{1 + \frac{4}{x^2}} + \frac{2}{1 + (2x)^2} \quad \textcircled{1}$$

$$= \frac{-2}{x^2(x^2 + 4)} + \frac{2}{1 + 4x^2} \quad \textcircled{1}$$

$$= \frac{-2}{x^2 + 4} + \frac{2}{1 + 4x^2} \quad \textcircled{1}$$

(b)

(i) $v = (2-x)^2$

$$v^2 = (2-x)^4$$

$$\frac{1}{2} v^2 = \frac{1}{2} (2-x)^4 \quad \textcircled{1}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2(2-x)^3 \cdot (-1)$$

$$\frac{dv}{dx} v = -2(2-x)^3 \quad \textcircled{1}$$

(ii) $\frac{dx}{dt} = (2-x)^2$

$$\frac{dt}{dx} = \frac{1}{(2-x)^2} = (2-x)^{-2}$$

$$t = \frac{(2-x)^{-1}}{-1 \times -1} + C \quad \textcircled{1}$$

when $t=0$, $x=0$

$$0 = 2^{-1} + C$$

$$C = -\frac{1}{2} \quad \textcircled{1}$$

$$t = \frac{1}{2-x} - \frac{1}{2}$$

$$x = \frac{4t+1}{2t+1} \quad \textcircled{1}$$

(ii) when $|v|=1$

$$1 = (2-x)^2$$

$$2-x = \pm 1 \quad \textcircled{1}$$

$$x = 1 \text{ m or } 3 \text{ m} \quad \textcircled{1}$$

c)
$$\begin{array}{c} 32 \text{ Players} \\ / \quad \backslash \\ 20 \text{ Australian} \quad 12 \text{ others} \end{array}$$

$$P(\text{all Aust}) = \frac{{}^{20}C_{13}}{{}^{32}C_{13}} \quad \text{--- (1)}$$

$$= 0.00022316$$

Question (7).

(a) (i) $\cos 2A = 1 - 2\sin^2 A \quad \text{--- (1)}$

(ii)
$$\int_0^\pi \sin^2 \frac{x}{4} dx$$

$$= \int_0^\pi \left\{ \frac{1}{2} - \frac{1}{2} \cos \frac{x}{2} \right\} dx \quad \text{--- (1)}$$

$$= \left[\frac{x}{2} - 1 \cdot \sin \frac{x}{2} \right]_0^\pi$$

$$= \frac{\pi}{2} - 1 - 0$$

$$= \frac{\pi}{2} - 1 \quad \text{--- (1)}$$

(b) $x = 4 \sin \frac{\pi}{2} t$

(i) $x = 4 \sin \frac{\pi}{2} t$

$$\left. \begin{aligned} \dot{x} &= 2\pi \cos \frac{\pi}{2} t \\ \ddot{x} &= -\pi^2 \sin \frac{\pi}{2} t \end{aligned} \right\} \quad \text{--- (1)}$$

$$\ddot{x} = -\frac{\pi^2}{4} x$$

$$\left. \begin{aligned} &\text{in form } \ddot{x} = -n^2 x \end{aligned} \right\} \quad \text{--- (1)}$$

which is SHM. $n = \frac{\pi}{2}$

(ii) Amplitude = 4
 Period = $\frac{2\pi}{\left(\frac{\pi}{2}\right)} \left(\frac{1}{2} \text{ each}\right)$
 $= 4 \quad \text{--- (1)}$

(iii) max speed. when $\cos \frac{\pi}{2} t = 1$
 $\ddot{x} = 2\pi \cos \frac{\pi}{2} t$
 $\dot{x}_{\text{max}} = 2\pi \text{ m/s} \quad \text{--- (1)}$

(c) $(5+2x)^{12} = \sum_{k=1}^{12} a_k x^k$

$$a_k = {}^{12}C_k 5^{12-k} 2^k$$

(ii) $a_{k+1} = {}^{12}C_{k+1} 5^{12-k-1} 2^{k+1}$

$$\therefore \frac{a_{k+1}}{a_k} = \frac{{}^{12}C_{k+1} 5^{11-k} 2^{k+1}}{{}^{12}C_k 5^{12-k} 2^k}$$

$$= \left[\frac{12!}{(k+1)!(12-k-1)!} \right] 2 \cdot \frac{1}{5} \quad \text{--- (1)}$$

$$= \frac{12! \cdot 2}{(k+1)!(12-k-1)!} \times \frac{k!(12-k)}{12! \cdot 5}$$

$$= \frac{2(12-k)}{5(k+1)} \quad \text{--- (2)}$$

$$= \frac{24-2k}{5k+5}$$

as required.